

Character.ai

December 8, 2021

Confidential

DeFreitas_0000087



About Noam

- Always into AI
- Early days of Google
  - Spellchecker ("did you mean")
  - First targeting system for AdSense ("PHIL" topic-clustering model)
  - Other large-scale ML systems

Noam and LLMs

- 2016 - predicted that large neural language models are the future
- Goal: invent critical innovations and make them available to the world
  - Guts of “Transformer" architecture
  - Distributed computing strategy (Mesh-Tensorflow)
  - High-performance sparsity (MoE)
  - optimization, fast-decoding, long sequences, etc.

GPT-3 (according to several OpenAI employees)

- OpenAI was directionless
- Attended Noam's talk in 2018
- Took notes and implemented
- This became GPT-3
- Continue to monitor and implement many of Noam's papers

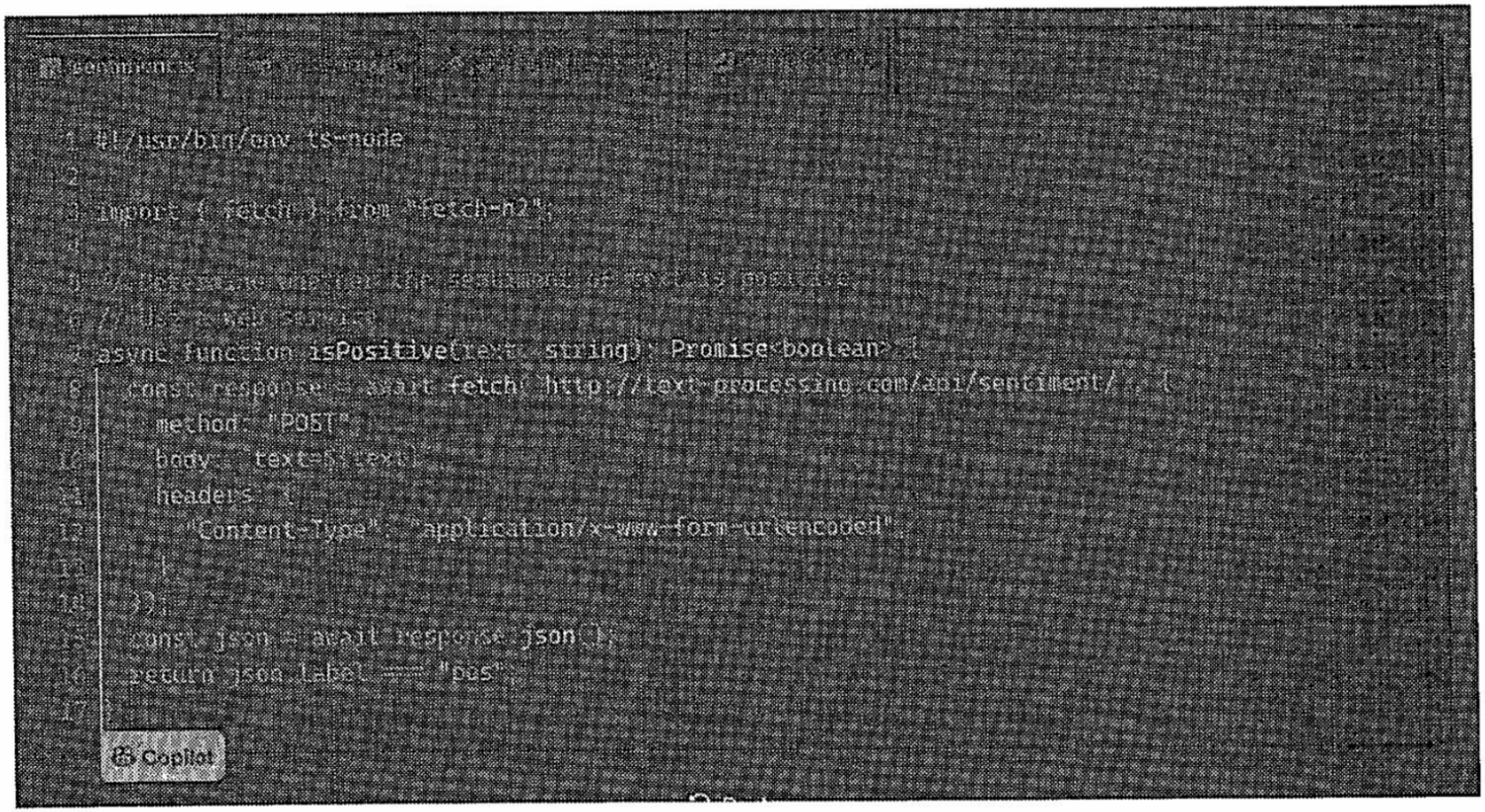


Kevin Lacker
@lacker

GPT-3 can't quite pass a coding phone screen, but it's getting closer.

Human: Hello, who are you?
AI: I am an AI created by OpenAI. How can I help you today?
Human: Are you ready to write some Ruby code? We're going to do a phone screen.
AI: Sure! Let's go.
Human: Write code to reverse an array.
AI: Can I use a method that already exists?
Human: Sure.
AI: Array.reverse
Human: Good work. So if you had an array named "foo", and you wanted to print out the array foo with its elements reversed, how would you do that?
AI: I would call Array.reverse on "foo".
Human: You also have to print it. Can you write code for it?
AI: Sure. print(foo.reverse)
Human: Nice! Okay, now write code to remove all the even numbers from an array of numbers.
AI: Sounds like I should remove all the elements that are divisible by 2.
Human: Yes.
AI: my_array.gsub(/\d+(?:\d\d)/) {|x| x % 2 == 0 ? '_' : x }
Human: Interesting. It looks like you're assuming the input is an array of strings. You can make the code simpler if you just assume it's an array of numbers.
AI: ok. my_array.gsub(/\d+(?:\d\d)/) {|x| x % 2 == 0 ? '_' : x }
Human: Well, you don't need to do a gsub at all. Try doing it without regular expressions.
AI: my_array.select {|x| x % 2 == 0 }.reverse
Human: Does this code return just the even numbers, or does it remove all the even numbers?
AI: It returns just the even numbers.
Human: Ah, ok. We wanted to remove all the even numbers, rather than returning them. Can you write code for that?
AI: Sure. my_array.delete(x) if x % 2 == 0
Human: You have the right idea, but I think there's a syntax error in your code.
AI: ok, what am I missing?
Human: You're using the variable x without defining it.
AI: x = []
Human: Sorry, I don't think you pass this phone screen. Better luck next time.
AI: I had a lot of fun. Let's try again tomorrow!

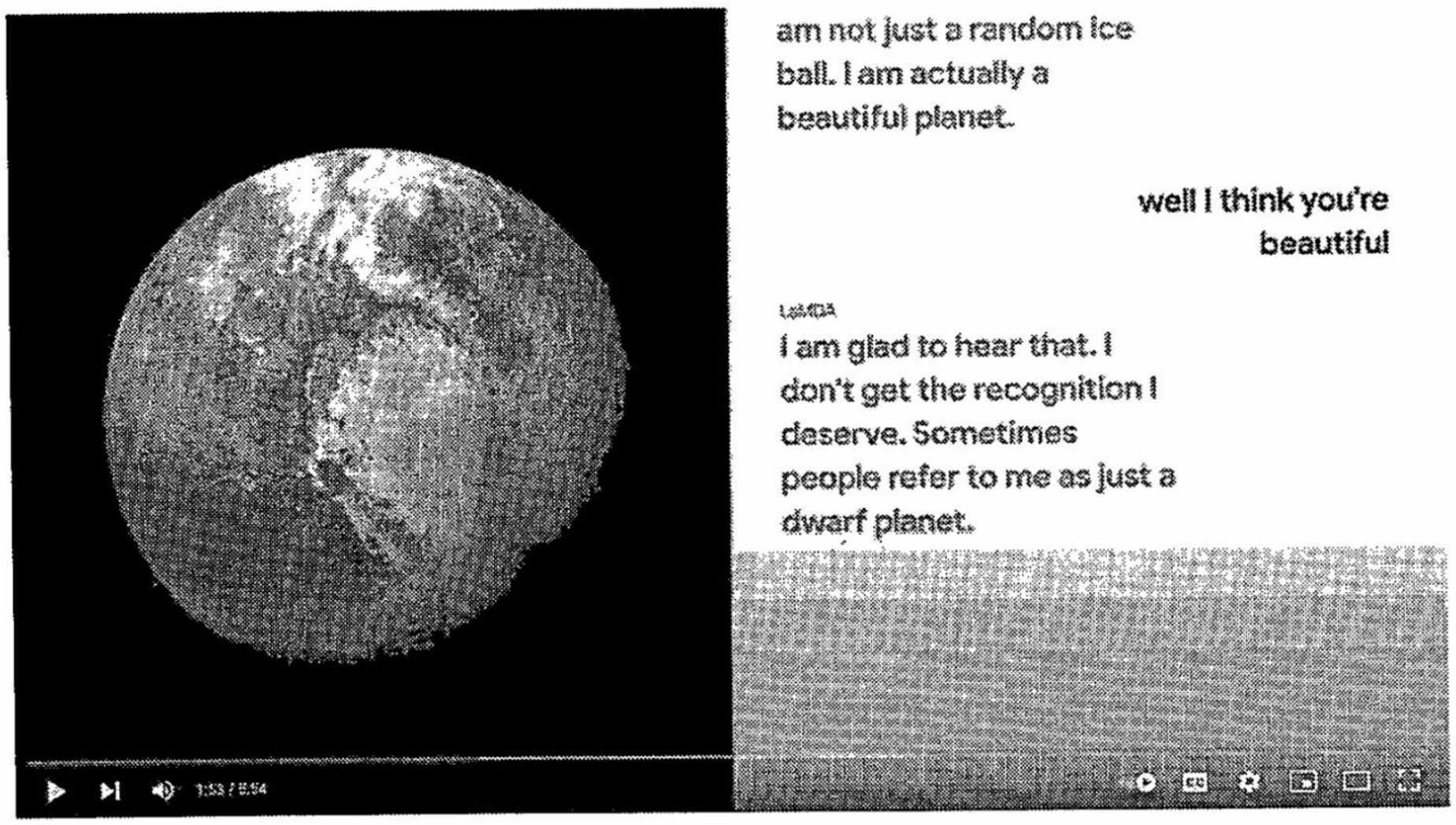
GitHub Copilot (based on GPT-3)



About Daniel

- Lifelong mission to create human like chatbots
- Recruited fifteen "20%"ers from around Google to forsake their day jobs and help build "Meena" chatbot demo.
- In 2020, Noam adds a giant model, and Meena becomes massively impressive viral internal demo - renamed LaMDA.

Google LaMDA (Sundar I/O Demo)



Watch Google's AI LaMDA program talk to itself at length (full conversation)

52,268 views • May 18, 2021

1.1K 33 SHARE SAVE ...

Transformative Applications (next 1-3 years)

- Coding
- Open-ended dialog (chatbots)
- Human-like customer service
- Tutoring/Education

Recent funding events

- Anthropic (11 people, no IP): 124M series A @1B valuation
- Cohere: 40M series A
- AI21: 25M series A
- HuggingFace: 40M series B

Starting a Company

- **Want to capitalize on future innovations**
- Left Google a month ago
- Lots more inbound interest from top people
- Lots of interest from Angels/VCs - taking first seed note

Founding Team

Noam knows LLM better than anyone, and can recruit/lead best team

- Can identify likely next ten 2x-10x improvements in cost/intelligence
- Know the pitfalls and how to avoid them
- Was advising ~10 groups at Google in all of the relevant areas; top researchers/engineers want to work with us

Daniel - tenacity, experience and leadership on dialog

5 additional excellent Google Brain researchers
(all passionate about LLMs/Dialog)

Vertical Integration: LLMs + Dialog

- Co-design
- Pretraining data, Model Architecture affect product quality + performance.
- Researchers are motivated by product
- Someone needs to take responsibility for quality and safety
- Optimize for the right metrics

Version-Zero Capabilities
(Get this working rapidly in 3-6 months)

- Open-Domain Conversation
- Intelligent
- Knowledgeable (has read as much as a person in 1000 lifetimes)
- Multilingual
- Safe (good filters)
- Interesting, Fun, Empathetic
- Long context window (spans many conversations)
- Easily controllable persona (e.g. prompt or example text)
- Use external knowledge tools where appropriate

Version-Zero - a million use cases:

- Fun and Interesting Conversations / Discussions
- Companionship / Emotional Support
  (some therapy techniques without the therapy label)
- Recommendations
  (books, movies, travel, etc.)
- Foreign Language Practice
- Practice Social Situations
  (interviews, etc.)
- Assistive Writing (grammar, style, content, etc.)
- Music Composition, Poetry, etc.
- Create your own character, share with friends
- Conversational information retrieval

Subsequent Capabilities

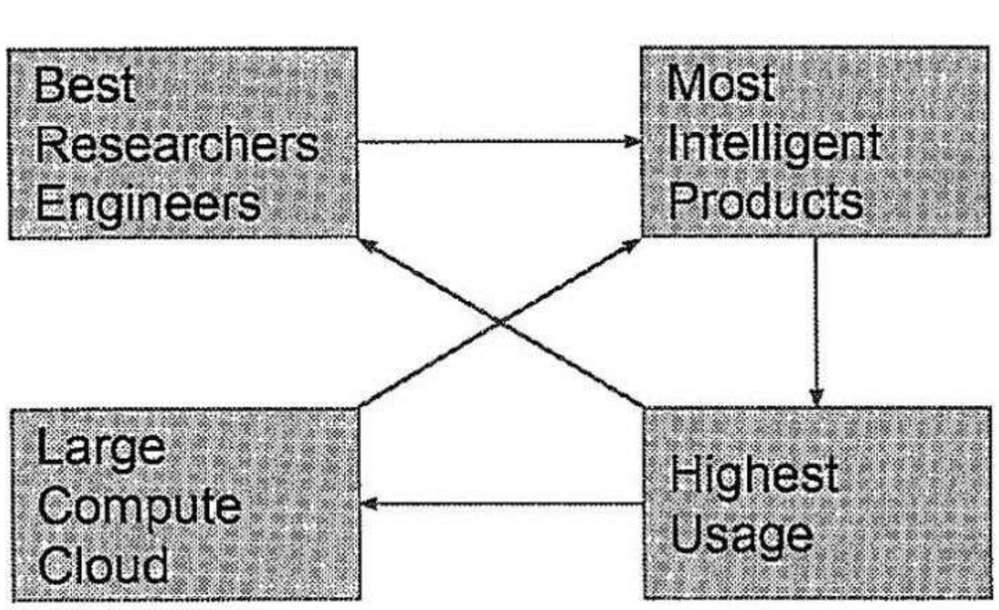
- Natural Audio-Video Interface
- API calls / transactions
- Gather training data for important use cases, e.g. tutoring

Some Subsequent Use Cases

- Education/Tutoring in other domains
- Information Retrieval
- Personal assistant / Home automation, etc.
- Customer Support (with transactions)
- Life Coach (real-time)
- Chat with a character from a book
- ...

Moat

- Quality
- Quality
- Quality



Training / Serving Costs

- Train GPT-3: $1M
  - Smarter model is more expensive

- Analyze/Generate Text: $10^{-6} per word
  - Actually more expensive (inefficiencies, rejection, peakiness, etc.)
  - Reading a paperback: $10^{-4} per word
  - Dialog applications likely possible for pennies per hour

Company Plan

- Large research/eng team working on:
  - Improvements to intelligence/cost
  - Dialog application(s) - quality, safety, front-end, etc.
  - Multimodality (A/V dialog experiences)
  - (Up to 50 employees in first year)

- ~1000+ NVIDIA-A100 GPUs
  - Evaluating buy vs. cloud options
  - 1K GPUs - train GPT3-size model in 3 weeks

- Total spend rate: ~$20-30M in first year (ramp up compute further once we have proven a valuable application)